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Project	Predict & Explain: Identifying Heart Disease Risk Factors
Subject	Data Science Mini Project – 2025–2026

Problem Statement

Background

Cardiovascular diseases (CVDs) account for approximately 17.9 million deaths per year worldwide, representing 32% of all global deaths (WHO, 2021). Heart disease — a subset of CVDs — is largely preventable if risk factors are identified early. However, clinical diagnosis often relies on expensive tests and specialist consultations that are not accessible to large populations, particularly in low- and middle-income regions.

Problem Definition

Given a set of routinely collected clinical measurements from a patient, can we accurately predict whether the patient has heart disease? Furthermore, which clinical features contribute most significantly to the risk — enabling healthcare providers to prioritise high-risk patients for further testing?

Objectives

- Build a binary classification model to predict heart disease from 13 clinical features.
- Perform Exploratory Data Analysis to identify patterns and correlations in the dataset.
- Compare multiple machine learning algorithms and select the best-performing model.
- Visualise feature importance to explain which risk factors matter most.
- Contribute to SDG Goal 3 by demonstrating data-driven approaches to healthcare.

Dataset

Attribute	Value
Source	UCI Machine Learning Repository – Cleveland Heart Disease
Records	303 patient samples
Features	13 clinical attributes (age, sex, cp, chol, thalach, etc.)
Target	Binary: 0 = No Disease, 1 = Heart Disease
Missing Values	15 cells across trestbps, chol, thalach (imputed)

SDG Alignment

This project directly supports **UN SDG Goal 3: Good Health and Well-being**, specifically Target 3.4 which aims to reduce by one-third premature mortality from non-communicable diseases through prevention and treatment. By developing an accessible, data-driven screening tool, this project contributes to democratising preventive healthcare diagnostics.